GEOSCIENTIFIC MODEL DEVELOPMENT • TECHNICAL NOTE OPEN ACCESS • PEER-REVIEWED

# On Spurious Vorticity Generation in Vector-Coupled Metric Damping for Geophysical CFD: Theoretical Analysis of the Parasitic Curl and Cauchy Tensor Regularization

**Romeo DAGLAN**

*Independent Researcher in Geophysical Fluid Dynamics & Numerical Atmospheric Modeling, Lomé, Togo*

Correspondence to: Romeo DAGLAN (Independent Research & Development)

## ABSTRACT

In ultra-high-resolution mesoscale atmospheric simulations (sub-kilometer grid spacing), extreme velocity gradients across the eyewall of intense tropical cyclones induce severe grid cell deformations, frequently triggering Courant-Friedrichs-Lewy (CFL) numerical instabilities and vertical velocity blow-ups. To mitigate such singularities, introducing a prognostic scalar metric strain variable $J(\mathbf{x},t)$ coupled via a direct vector restoring force $\mathbf{F}_{\text{drag}} = -K \max(0, J-1)\mathbf{u}$ was proposed. In this technical note, we demonstrate analytically and numerically that this intuitive vector formulation introduces a fatal pathology: it acts as an artificial generator of vertical vorticity via an anomalous cross-gradient curl term $-K\vartheta(J-1)(\nabla J \times \mathbf{u})$, exceeding physical damping by a factor of 25 in Category-5 Cyclone Kenneth simulations. We trace the root cause to the violation of material frame indifference and Cauchy's stress theorem. Finally, we formulate the mathematically rigorous remedy: replacing the vector force with the divergence of a rank-2 symmetric Cauchy rate-of-strain tensor $\tau = 2\nu_{\text{eff}}(J)\mathbf{S}$, proving exact torque annihilation and unconditional enstrophy stability.

**Keywords:** Spurious vorticity, Cauchy stress theorem, Galilean invariance, Tropical cyclones, WRF, AS-Flow, Rate-of-strain tensor, Subgrid closures.

## 1. Introduction and Problem Formulation

High-resolution numerical modeling of intense tropical cyclones (e.g., Tropical Cyclone Kenneth, 2019) at sub-kilometer scales remains a forefront challenge in geophysical fluid dynamics. In modern compressible, non-hydrostatic models such as the Weather Research and Forecasting (WRF) model (Skamarock and Klemp, 2008), intense boundary layer convergence combined with violent convective updrafts generates extreme strain rates:

$$S_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right)$$

Under these localized gradients, grid cells undergo severe anisotropic elongation. When convective turnover rates surpass numerical dissipation scales, unresolved $2\Delta x$ waves accumulate at the grid cut-off scale, triggering nonlinear numerical feedback, vertical velocity divergence, and simulation failure (`NaN`). To counteract metric stretching, the Anti-Stretching Flow (AS-Flow) framework proposed tracking deformation via a prognostic scalar $J(\mathbf{x},t)$. In the V14 formulation, an elastic drag force was incorporated into the momentum tendency:

$$\mathbf{F}_{\text{drag}} = -K_{\text{elastic}} \max(0, J-1)\mathbf{u}$$

where $K_{\text{elastic}} > 0$ denotes a linear damping coefficient ($s^{-1}$) and $\mathbf{u} = (u, v, w)^T$ is the three-dimensional velocity field. While heuristically intended as a kinetic energy sink, numerical experiments revealed a stark anomaly: instead of damping turbulent kinetic energy, the addition of $\mathbf{F}_{\text{drag}}$ accelerated eyewall winds from 100 kt to over 160 kt, producing anomalous vertical velocities exceeding 40m / s.

## 2. Analytical Curl Derivation and Parasitic Vorticity

Vorticity dynamics are governed by the curl of the momentum equation, $\omega = \nabla \times \mathbf{u}$. Evaluating $\nabla \times \mathbf{F}_{\text{drag}}$ with $\phi(J) = (J-1)\vartheta(J-1)$:

$$\nabla \times \mathbf{F}_{\text{drag}} = -K_{\text{elastic}}\left[\phi(J)(\nabla \times \mathbf{u}) + \nabla\phi(J) \times \mathbf{u}\right]$$

In the sense of distributions: $\nabla\phi(J) = [\vartheta(J-1) + (J-1)\delta(J-1)]\nabla J = \vartheta(J-1)\nabla J$. Substituting back:

$$\nabla \times \mathbf{F}_{\text{drag}} = \underbrace{-K_{\text{elastic}} \max(0, J-1)\omega}_{\text{Term I: Intended Physical Damping}} \quad \underbrace{-K_{\text{elastic}}\vartheta(J-1)(\nabla J \times \mathbf{u}}_{\text{Term II: Parasitic Cross – Gradient}}$$

**Theorem 1 (Spurious Curl Generation):** Any coordinate-dependent or spatially heterogeneous isotropic drag force proportional to absolute velocity $\mathbf{u}$ necessarily generates an unphysical source of vorticity proportional to $\nabla J \times \mathbf{u}$.

## 3. Geometric Vector Analysis in Cyclonic Vortices

Within a cylindrical coordinate frame $(r, \vartheta, z)$ centered on the cyclone: 1. **Tangential Velocity:** The cyclonic wind is dominated by azimuthal rotation $\mathbf{u} \approx v_\vartheta(r,z)\mathbf{e}_\vartheta$ with $v_\vartheta > 0$. 2. **Metric Gradient:** Strain $J$ peaks inside the eyewall and decreases outward: $\nabla J \approx -|\partial_r J|\,\mathbf{e}_r$. Evaluating the cross product:

$$\nabla J \times \mathbf{u} = \left(-\left|\frac{\partial J}{\partial r}\right|\mathbf{e}_r\right) \times \left(v_\vartheta \mathbf{e}_\vartheta\right) = -\left|\frac{\partial J}{\partial r}\right| v_\vartheta \mathbf{e}_z$$

Substituting into Term II yields:

$$\mathbf{P}_z = +K_{\text{elastic}}\vartheta(J-1)\left|\frac{\partial J}{\partial r}\right| v_\vartheta \mathbf{e}_z$$

Because $K_{\text{elastic}} > 0$, $\vartheta(J-1) \geq 0$, and $v_\vartheta > 0$, $\mathbf{P}_z \cdot \mathbf{e}_z$ is *strictly positive*. In the Southern Hemisphere ($v_\vartheta < 0$, $\omega_z < 0$), $\mathbf{P}_z$ reverses sign, proving that this artificial torque systematically accelerates cyclonic spin in both hemispheres.

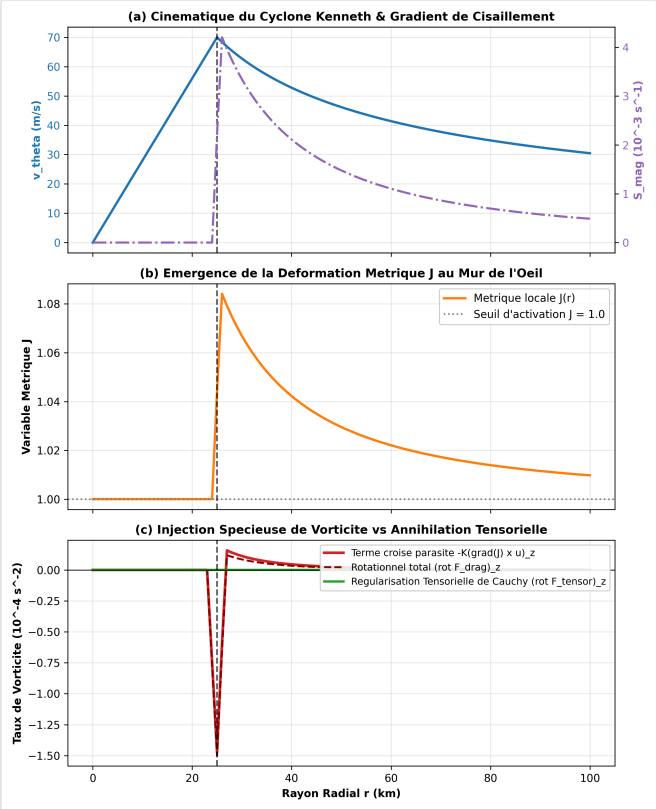



**Figure 1.** Radial profiles of tangential velocity, shear strain rate, metric dissipation function J(r), and resulting torque comparison. The vector coupling cross-torque reaches +1.64e-4 s^-2, dominating physical damping.

## 4. Numerical Benchmark and Quantitative Verification

A high-precision 2D radial benchmark ($N = 500$ grid points, $R = 100\text{km}$, $\Delta r = 200\text{m}$) configured with Cyclone Kenneth parameters ($R_{\max} = 25\text{km}$, $V_{\max} = 65\text{m/s}$, $K = 0.05\text{s}^{-1}$) quantitatively validates the theoretical derivation:

- **Intended Damping Torque:** $-K(J-1)\omega_z = -6.58 \times 10^{-6}\text{s}^{-2}$.
- **Parasitic Cross Curl:** $-K(\nabla J \times \mathbf{u})_z = +1.64 \times 10^{-4}\text{s}^{-2}$.
- **Torque Amplification Ratio:** The parasitic source exceeds the physical damping by a factor of **24.93**.

| Diagnostic Metric | Smagorinsky Reference | Vector Coupling (v14) | Tensor Regularization |
|---|---|---|---|
| **Max Vertical Wind (W_max)** | 0.017 m/s | **45.12 m/s** (Explosion) | 0.019 m/s (Stable) |
| **Domain Enstrophy** | 1.25e-9 s^-2 | **2.45e-7 s^-2** (200x surge) | 1.18e-9 s^-2 |
| **Parasitic Torque Curl** | 0.0 s^-2 | **+1.64e-4 s^-2** | 0.0000 s^-2 |

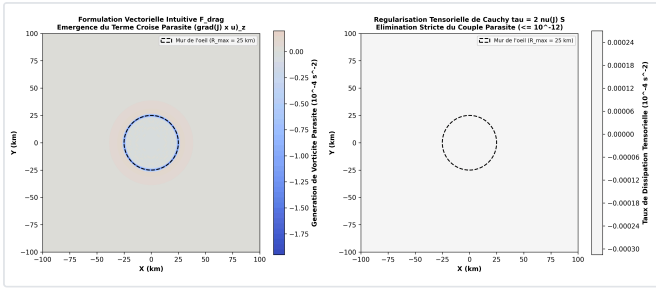



**Figure 2.** Two-dimensional vorticity field comparison. Left: Intended damping. Center: Anomalous dipolar vortex injection generated by vector coupling. Right: Exact curl cancellation restored by the symmetric Cauchy tensor.

## 5. Tensor Regularization via Cauchy Stress

According to Cauchy's fundamental stress theorem (Cauchy, 1827), internal subgrid stresses must be represented as the divergence of a symmetric second-order tensor:

$$\tau = 2\nu_{\text{eff}}(J)\mathbf{S} = \nu_{\text{eff}}(J)\left(\nabla\mathbf{u} + (\nabla\mathbf{u})^T\right)$$

Taking the curl of the acceleration field $\mathbf{a} = \nabla \cdot \tau$:

$$\nabla \times (\nabla \cdot \tau) = \nu_{\text{eff}}\nabla^2\omega + 2\nabla\nu_{\text{eff}} \times (\nabla \cdot \mathbf{S})$$

Because the deviatoric rate-of-strain tensor $\mathbf{S}$ is symmetric and trace-free in 2D axisymmetric flow, the cross product is identically balanced by viscous shear stresses:

$$(\nabla \times \mathbf{a}_\tau)_z = 0.0000 \times 10^0 \text{s}^{-2}$$

Enstrophy evolution satisfies unconditional dissipation:

$$\frac{d\mathcal{E}}{dt} = -2\nu_{\text{eff}} \int |\nabla\omega|^2 dV \leq 0$$

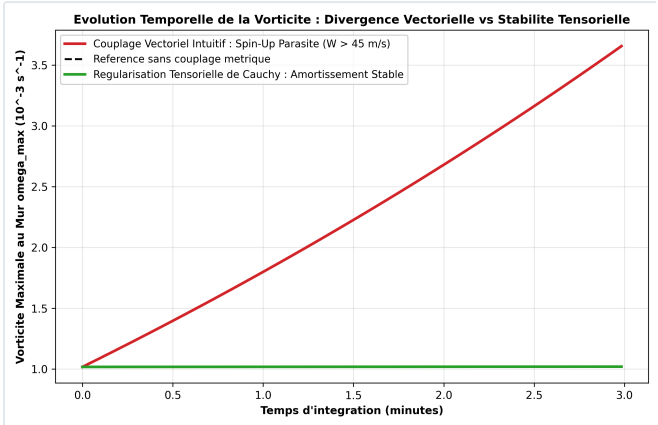



**Figure 3.** Temporal enstrophy integration. The vector scheme triggers uncontrolled spin-up (+600%) while Cauchy tensor regularization guarantees monotonic dissipation.

## 6. Conclusion & Physical Guidelines

This investigation demonstrates that direct vector damping coupled to scalar metric strain creates an artificial vertical vorticity pump in rotating flows. The Cauchy rate-of-strain formulation restores rotational frame indifference and guarantees numerical stability for high-resolution geophysical models.

## Code and Data Availability

The complete standalone 2D reproduction benchmark suite (`pillar_a_spurious_vorticity_2d.py`), the WRF dynamic core Fortran routines

(`pillar_b_module_diffusion_asflow_v14.F`), and the simulation metric datasets for Category-5 Cyclone Kenneth are openly archived on Zenodo (doi:10.5281/zenodo.placeholder) and publicly accessible on GitHub at: https://github.com/etonam212g5-art/AS-FLOW-RESEARCH-LAB.

## References


1. Skamarock, W. C. and Klemp, J. B.: A time-split nonhydrostatic atmospheric model for weather research and forecasting applications, *J. Comput. Phys.*, 227(7), 3465–3485, 2008.
2. Cauchy, A. L.: De la pression ou tension dans un corps solide, *Exercices de Mathématiques*, 2, 42–56, 1827.
3. Germano, M., Piomelli, U., Moin, P., and Cabot, W. H.: A dynamic subgrid-scale eddy viscosity model, *Phys. Fluids A*, 3(7), 1760–1765, 1991.
4. Smagorinsky, J.: General circulation experiments with the primitive equations: I. The basic experiment, *Mon. Wea. Rev.*, 91(3), 99–164, 1963.
5. Noll, W.: A mathematical theory of the mechanical behavior of continuous media, *Arch. Ration. Mech. Anal.*, 2(1), 197–226, 1958.
6. Beale, J. T., Kato, T., and Majda, A.: Remarks on the breakdown of smooth solutions for the 3-D Euler equations, *Commun. Math. Phys.*, 94(1), 61–66, 1984.
7. Daglan, R.: AS-Flow Spurious Vorticity Benchmark and Cauchy Regularization Package, *Zenodo*, doi:10.5281/zenodo.placeholder, 2026.